

Tirdesh Pettugani

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EDUCATION

Northeastern University, Boston, MA

December 2025

Master of Science in Information Systems

GPA: 3.7/4

Related Courses: Knowledge Graphs with LLM/Graph DB, WebDev Tools, Program Structures and Algorithms

WORK EXPERIENCE

DMSB AI Strategic Hub, Boston, USA

Software Developer & AI Engineer

08/2025 – Current

- Designed and deployed dashlab.io as DASH Lab's public-facing platform, showcasing research innovations that generated partnership discussions and attracted inquiries from 5+ potential investors and industry stakeholders.
- Enriched 70K+ alumni records by engineering automated web scraping pipelines and third-party API integrations, transforming disparate data sources into structured datasets for targeted outreach.
- Launched AI-driven educational tools, including EssayBot (streamlined essay grading using Gemma models deployed on A6000 infrastructure) and PresBot (real-time presentation coaching powered by voice cloning ML models), transforming manual processes into scalable, personalized feedback systems.

Wave Life Sciences, Lexington, USA

IT Intern

01/2025 – 06/2025

- Architected an AI-assisted knowledge retrieval system using Gemini models and Vertex AI with RAG capabilities, developing a Python FastAPI backend to process research documents and power an internal chatbot for intelligent query responses.
- Developed interactive dashboards integrating multiple service APIs for data visualization and automated reporting, reducing manual data compilation time by 60% and streamlining cross-team access to research information.

Commvault Systems, Hyderabad, India

Software Development Engineer

01/2021 – 08/2023

- Contributed to 7+ revenue-generating projects over 4 years, delivering key features and enhancements that earned spot bonuses and company recognition for technical excellence.
- Designed and engineered responsive React components from Figma mockups, collaborating with design and backend teams to ensure seamless UI/UX and efficient data integration via GraphQL and REST APIs.
- Implemented a Selenium-Python automation framework to validate backup and recovery functionality for GitHub and Azure DevOps repositories, boosting verification efficiency by 85% and reducing validation time by 50%.
- Led 3 full-cycle projects from design to deployment, conducted code reviews to maintain quality standards, and mentored 5 junior engineers in Python development and best practices.
- Resolved 200+ customer escalations through systematic debugging and root cause analysis, improving product reliability and customer satisfaction.

Associate Engineer

01/2020 – 01/2021

- Won internal hackathon (Create 2020) by developing a Chrome extension using Node.js and webhooks that enhanced web navigation workflows for internal tools.
- Managed Hadoop distributions and MongoDB deployments, conducting bare metal recovery across UNIX systems (Rocky Linux, RedHat, CentOS, Ubuntu) to ensure system reliability and data integrity.

TECHNICAL SKILLS

Languages & Frameworks	Python, Java, JavaScript, TypeScript, React, Node.js (Express.js), Next.js, Flask
Databases & Big Data	MongoDB, MySQL, Postgres, SQL Server, Hive, HBase, HDFS, Neo4j, Aura
AI/ML & Data Science	TensorFlow, Keras, NumPy, Pandas, OpenCV, LangChain, Llama4, OpenAI
Cloud & DevOps	Docker, Kubernetes, Azure, AWS, GCP, OCI, Linux
Web & APIs	HTML, CSS, REST API, GraphQL, TCP/IP, OSI Model

PROJECTS

LearnX, Northeastern University | youtu.be/lk7-5Qedr1o | *iOS, SwiftUI, Firebase*

- Developed an iOS learning platform for mastering Data Structures and Algorithms with structured modules, interactive quizzes, real-time progress tracking, and community-driven discussion forums for mobile-first learning.

MagicLetter, Personal Project | magicletter.vercel.app | *Vite React, Firebase, Shadcn-ui, Tailwind CSS*

- Built a scalable cover letter generation application using Firebase and AI providers (Cohere, OpenAI, Claude) with real-time preview, multiple templates, and easy customization for streamlined job application workflows.

Find A Roomie, Northeastern University | find-a-roomie.vercel.app | *MERN stack*

- Created a web application streamlining accommodation search for international students with integrated real-time messaging, blog features, and collaborative tools to facilitate roommate connections.